1 A worked example: Kepler and Newton

We work through Kepler's laws, the theories of Mars that preceded the ellipse, and what Newton adds. What a law says depends on the data language (Section 1.2); in a proof-requiring posterior over the historical theories of Mars the data decide compatibility and, on noise-free data, the language of the law texts decides among compatible theories (Section 1.4); a successor's gain need not lie in a better fit to the old data (Section 1.5). The data are computed with the ERFA library, so *every positional error in this section is against the plan94 ephemeris (moon98 for the Moon), not against the sky.* Appendix A has proofs and tables; the code is in the accompanying repository.

1.1 The language and the data

Definition 1.1 (The Kepler languages and theories)**.**

(i) *Data languages.* Sorts bodies and reals; body constants Sun, Me, Ve, Ea, Ma, Ju, Sa, Mo; a predicate Planet. The *heliocentric language* $\mathcal{L}_{\text{hel}}$ has the heliocentric ecliptic longitude, latitude and distance $\theta(b,t), \beta(b,t), r(b,t)$ of body b at time t (for the Moon, geocentric); its direction-only part $\mathcal{L}_{\text{dir}}$ omits r. The *geocentric language* $\mathcal{L}_{\text{geo}}$ has $\theta_g(b,t), \beta_g(b,t)$, giving the direction $\text{dir}_g(b,t) \in S^2$: a modern idealisation of Tycho's data.

(ii) *Background.* B: the axioms of real closed fields with sin, cos, π and enough true analysis to prove the numerical bounds used, a sound decidable fragment of the undecidable theory of the reals with sin (Appendix A.1); each result names the true sentences it needs.

(iii) *Kepler's first two laws* K12, in the theoretical vocabulary of heliocentric positions $\mathbf{r}(b,t) \in \mathbb{R}^3$ and the elements $a, e, i, \Omega, \varpi, M_0, P$ (functions from bodies to reals: semi-major axis, eccentricity, inclination, node, longitude of perihelion, mean anomaly at the epoch, period): for every planet b and every t, with $M := M_0(b) + 2\pi t/P(b)$, there is ψ with $\psi - e(b)\sin\psi = M$ and $\mathbf{r}(b,t) = \text{Rot}(b)(a(b)(\cos\psi - e(b)),\ a(b)\sqrt{1-e(b)^2}\sin\psi,\ 0)$, where $\text{Rot}(b)$ is the rotation given by $i(b), \Omega(b), \varpi(b)$; moreover $0 \le e(b) < 1$, $a(b), P(b) > 0$ and $\mathbf{r}(\text{Sun}, t) = 0$. Over $\mathcal{L}_{\text{geo}}$, K12 also contains the frame convention $i(\text{Ea}) = 0$ and the *bridge*: $\text{dir}_g(b,t)$ is the unit vector of $\mathbf{r}(b,t) - \mathbf{r}(\text{Ea},t)$ if this is nonzero, and the north ecliptic pole $\mathbf{z}$ otherwise.

(iv) *Kepler's third law* K3: $P(b)^2 a(b')^3 = P(b')^2 a(b)^3$ for all planets b, b'.

(v) *Elements* (initial conditions): numerals, or intervals, for the elements of a body. *Data*: true interval sentences $\text{Near}(b,t,\mathbf{v},\delta)$, "the direction of b at t lies within angle δ of $\mathbf{v}$" (Near_g for geocentric directions), with numerals for $t, \mathbf{v}, \delta$, and, where available, $r(b,t) \in [r_1, r_2]$. *Questions* (**??**): covers by such sentences ("where is Mars at t, to within $2'$?"), and structural questions, which the laws answer without elements (is the orbit planar?).

Laws alone answer no position question;[1] laws with elements do. With the elements in the antecedents of conditional questions the laws alone answer them, and in the model of **??** no steeper MAP then contains elements: this is where laws and initial conditions part formally.

1.2 What the laws say, and in which language

Proposition 1.2 (No content in heliocentric directions)**.** *For every B that contains the axioms of real closed fields,* K12 *and* K12 ∪ K3 *have the same $\mathcal{L}_{\text{dir}}$-consequences.*

[1] Even the Ramsey sentence of K12 ∪ K3, with any background true in $\mathbb{R}$, entails (in second-order semantic consequence) no datum $\text{Near}(b,t,\mathbf{v},\delta)$ with $\delta < \pi$ (Proposition A.1).

In a geocentric language K3 has content: K12 ∪ K3, with two true sentences of the real field, proves that Mars is not permanently in opposition, and K12 with any background true in $\mathbb{R}$ does not (Proposition A.2). The natural form of that content is this:

Proposition 1.3 (What K3 does in geocentric directions)**.** *Let* Num^- *be numerals for all elements of the Earth and Mars except* $a(\mathrm{Ma})$*, with* $a(\mathrm{Ea}) = 1$*. Let* $\mathbf{F}(t)$ *be Mars's heliocentric position computed from* Num^- *with semi-major axis* 1*, so that* $\mathbf{r}(\mathrm{Ma}, t) = \rho\,\mathbf{F}(t)$ *with* $\rho := a(\mathrm{Ma})$*, and let* $\mathbf{u}(t)$ *be the unit vector of* $\mathbf{r}(\mathrm{Ea}, t)$*.*

(a) $\mathrm{K12} \cup \mathrm{K3} \cup \mathrm{Num}^-$ *fixes* $\rho = (P(\mathrm{Ma})/P(\mathrm{Ea}))^{2/3}$ *and hence every geocentric direction of Mars: it proves* $\mathrm{Near}_g(\mathrm{Ma}, t, \mathbf{v}, \delta)$ *whenever the computed direction lies within* δ *of* $\mathbf{v}$ *and* B *proves the numerical bound involved.*

(b) $\mathrm{K12} \cup \mathrm{Num}^-$ *leaves* ρ *free. At every* t *at which* $\mathbf{r}(\mathrm{Ea}, t)$ *and* $\mathbf{F}(t)$ *are not parallel, the geocentric directions of Mars at* t *in the models of* $\mathrm{K12} \cup \mathrm{Num}^-$ *over* $\mathbb{R}$ *(with* ρ *ranging over* $(0, \infty)$*) form the open great-circle arc from* $-\mathbf{u}(t)$*, the direction of the Sun, to* $\mathbf{F}(t)/|\mathbf{F}(t)|$*; its length* $\alpha(t) \in (0, \pi)$ *is the angle between these two directions. So for every* B *true in* $\mathbb{R}$*,* $B \cup \mathrm{K12} \cup \mathrm{Num}^-$ *proves no datum* $\mathrm{Near}_g(\mathrm{Ma}, t, \mathbf{v}, \delta)$ *with* $\delta < \alpha(t)/2$*.*

A computation on the daily geocentric longitudes of Mars over 1580–1600 puts what K3 saves on the elements of Mars at 11.3, 12.3, 13.4 and 14.7 bits at $\delta = 8'$, $4'$, $2'$, $1'$, against a text of 375 bits in our arithmetic syntax and 165 or 115 bits in two Polish-notation codes; so in none of these codes does compression alone select K3 at Kepler's precision, an extrapolation from Mars to the other planets that is conditional on the code (Example A.4).

K3 is thus a law about initial conditions, worth what it saves on the elements of the bodies it relates. This depends on Kepler's kinematic parametrisation, with the period and the semi-major axis both free; in a state-space parametrisation K3 fixes the mean motion from the energy, a different question, and in Newton's theory the third law, with its mass correction, follows from the one law that does all the other work and costs nothing extra.

1.3 How wrong Kepler's laws are

Kepler's laws are approximately true, with an error that grows with the domain: one fixed least-squares ellipse misses Mars by $1.6'$–$1.9'$ over 12–30 years and $51.8'$ over 2000 years, and the Moon by 1.1° within a month (Table 2). The third-law "constant" $q := (2\pi/P)^2 a^3/k^2$ (k Gauss's constant) differs between planets at the 10^{-3} level, yet an exact K3 calibrated on the Earth misplaces Jupiter and Saturn geocentrically by at most $0.23'$ and $0.19'$ over 1580–1600, well below Tycho's precision. By the accuracy quoted in the documentation of the ERFA routines `eraPlan94` and `eraMoon98` (which pyerfa wraps), plan94's own error in the heliocentric longitude of Mars over 1580–1600 is at most about $0.43'$ (Appendix A.2).

1.4 The proof-requiring posterior over the historical Mars pool

The pool and the data. We take a hand-picked pool of five forms of theory for Mars, as heliocentric reconstructions: the circle about the Sun (2 elements), the eccentric circle (4), Ptolemy's bisected equant (4), Kepler's vicarious hypothesis (5) and the ellipse K12 (4) (Appendix A.3). The data are the 7306 daily plan94 heliocentric longitudes v_i of Mars over 1580.0–1600.0, as sentences "$\theta(t_i)$ lies within δ of v_i": the noise-free case of the guaranteed-tolerance regime of **??**. A form proves every datum, for suitable elements, iff δ is at least its *minimax error*, the least over

Table 1: The proof-requiring posterior over the Mars pool: law-text lengths and Occam factors (bits; "–": δ below the minimax error; all values in App. A.3). A form's weight is $\propto 2^{-(\text{text+Occam factor})}$. K3's text: 375 bits in S1 and S3, 450 in S2.

	circle	eccentric	equant	vicarious	ellipse
text, S1	155	445	990	990	545
text, S2	186	342	450	468	654
text, S3	155	445	990	990	260
$\mathrm{Occ}_{\mathsf{F}}(720')$	20.93	20.24	21.20	22.77	21.18
$\mathrm{Occ}_{\mathsf{F}}(30')$	–	45.38	40.35	42.83	39.65
$\mathrm{Occ}_{\mathsf{F}}(9')$	–	–	57.34	51.78	46.81
$\mathrm{Occ}_{\mathsf{F}}(1.6')$	–	–	–	74.54	59.93

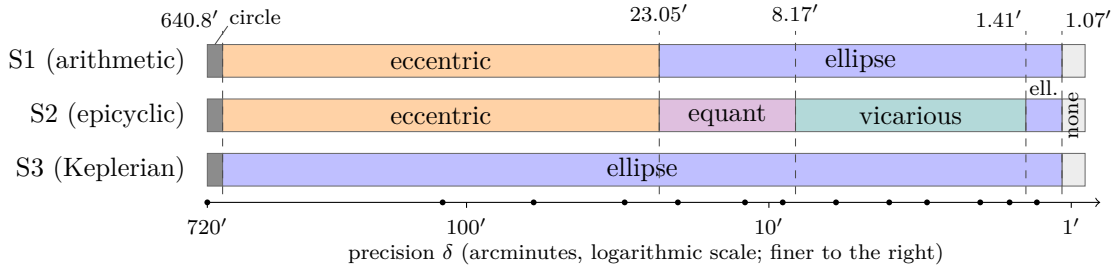


Figure 1: The MAP form of the proof-requiring posterior over the Mars pool against the precision δ of the data, in the three syntaxes, in the noise-free common-tolerance regime. Dashed lines: the minimax errors of Table 3, below which a form is incompatible with the data; below $1.07'$ no form in the pool is compatible. Dots: the precisions at which the posterior was computed; at each of them the runner-up has at most $2^{-12.4}$ of the MAP's weight. Between dots each switch is drawn at the minimax error of the form that drops out; that form stops being the MAP somewhere between this line and the nearest dot to its left.

elements of its largest error: $640.79'$, $23.05'$, $8.17'$, $1.41'$ and $1.07'$ in this order (upper bounds witnessed by elements, only the circle's proved optimal; Table 3). The equant's $8'$ remains against an exact ellipse, close to the famous eight-minute discrepancy of Kepler's *Astronomia Nova* (Kepler, 1992); the $0.34'$ between the vicarious hypothesis and the ellipse is of the size of plan94's error, so against the sky their ordering rests on the errors against an exact ellipse.

The posterior. A theory is a law text in one of three syntaxes plus fixed-length 64-bit numerals for its elements; the proof-requiring rule of M_A (**??**), with the universal machine replaced by this code, gives it weight $2^{-\text{length}}$ if it proves every datum and 0 otherwise. Summing over numerals, form F has weight proportional to $2^{-(|\mathrm{LAW}_{\mathsf{F}}|_S + \mathrm{Occ}_{\mathsf{F}}(\delta))}$, with $|\mathrm{LAW}_{\mathsf{F}}|_S$ the length of its law text in syntax S and $\mathrm{Occ}_{\mathsf{F}}(\delta) := -\log(\mathrm{Vol}_{\mathsf{F}}(\delta)/\prod \text{ranges})$ its *Occam factor*, $\mathrm{Vol}_{\mathsf{F}}(\delta)$ being the volume of the elements with which F proves all the data (the term is MacKay's (MacKay, 1992); Jefferys and Berger, 1992 read Bayes factors this way for Mercury's perihelion). The syntaxes are *S1*, arithmetic $(+, -, \cdot, /, \sin, \cos, \mathrm{atan2}, \sqrt{\ })$, the numerals $0, 1, 2$, quantifiers); *S2*, epicyclic (S1 with primitives for unit vectors, directions, and the point where a ray meets a circle); and *S3*, Keplerian (S1 with the true anomaly as a function of the eccentricity and the mean anomaly).

What the data decide and what the syntax decides (Table 1 and Figure 1). (i) *The data decide compatibility:* as δ falls the forms drop out in the historical order, in every syntax. (ii) *Among compatible forms the data contribute only the Occam factors, which on their own point weakly to the true form:* with a uniform prior over the five forms and no text prior, the MAP is the ellipse at every $\delta \leq 60'$, by 0.20 to 14.61 bits. (iii) *The text prior decides the rest:* law-text differences are 100–835 bits for every pair of forms except the equant and the vicarious hypothesis, whose texts are 18 bits apart in the epicyclic syntax, where the Occam factors move the margin by up to 5.6 bits, so the text still decides, and of equal length in S1 and S3, where both lose to the ellipse by more than 400 bits. So in this regime the syntax sets the sequence of MAP forms: with epicyclic primitives every form in the historical order, switching only when a form becomes incompatible; with plain arithmetic the eccentric, then the ellipse from $23'$; with a Keplerian primitive, the circle, then the ellipse from $641'$. A variable-length numeral code, Polish notation or a natural form outside the pool changes no MAP form (Appendix A.3).

"The simplest theory compatible with the data" is thus, in this regime, relative to the language of the law texts. A semanticist will call this relativity an artefact of a prior on texts. Our computation is mostly semantic: compatibility is a condition on a region of element space and the Occam factor is its measure, so only the law-text prior is syntactic; with a presentation-free prior, uniform over the five families and by volume on their elements, the data alone select the ellipse at every $\delta \leq 60'$, by small margins ((ii) above). But a prior over infinitely many families of models needs an enumeration of them, which is a presentation, and a volume prior depends on the parametrisation (halving the eccentricity ranges moves the vicarious hypothesis by one bit against the four-element forms), so the relativity is relocated rather than removed. The universal prior bounds this relativity by the cost $K(\tau)$ of a faithful translation (**??**), but that cost, here the definitions of the extra primitives in S1, is a few hundred bits (an estimate), as large as the text differences. Best-system accounts of laws face the same relativity (Lewis requires natural properties (Lewis, 1983); Cohen and Callender accept relativity to a chosen set of kinds (Cohen and Callender, 2009)), by analogy only: best systems are a metaphysics of laws, SAI is inference from finite data. And the noise-free regime is special: a form proves all the data or none, so the power of the data to discriminate among compatible forms does not grow with their number.

Graded fit and distances. With graded fit (a Gaussian error model with $\sigma = 2'$, roughly Tycho's precision) the data decide every step within tens of observations, except the epicyclic syntax's choice between the vicarious hypothesis and the ellipse on longitudes, which needs about 5,300 (Example A.3). With distances within 1% as data, compatibility alone eliminates the vicarious hypothesis below a longitude tolerance that is at most $5.12'$, in every syntax, which reproduces the abandonment of circles, not Kepler's step from an oval to the ellipse (Table 4).

1.5 What Newton adds

Newton's theory adds masses, the constant G and accelerations, with the law $m_b\ddot{\mathbf{r}}_b = \sum_{b' \neq b} G m_b m_{b'} (\mathbf{r}_{b'} - \mathbf{r}_b)/|\mathbf{r}_{b'} - \mathbf{r}_b|^3$, in the same data language; masses are theoretical in Sneed's sense (Sneed, 1971; Balzer et al., 1987). We see five points, gains of Newtonian mechanics over about 150 years, not of Newton alone (Newton accounted for about half of the motion of the lunar apsides, Clairaut for the rest in 1749 (Nauenberg, 2015; Bodenmann, 2010)). (1) *No gain on the old data:* on 20 years of longitudes at Kepler's precision the masses buy nothing measurable, so we think an SAI posterior on such data alone would not prefer Newton's theory. (2) *Unification,* where the gain lies: planets, Moon, falling bodies, tides and comets under one law

(some with more base vocabulary), and the value $q = 1 + m/M$ of the third-law constant, which Jupiter's fitted q matches. (3) *Corrections in the old domain:* in the classical decomposition (cf. Clemence, 1947), Newtonian planetary perturbations give about $531''$ of the fixed-frame advance of Mercury's perihelion, about $574''$ per century ($575.4''$ in our fits), and general relativity the remaining $43''$ or so; Laplace explained the great inequality of Jupiter and Saturn (Wilson, 1985). (4) *New questions:* in the old language, where Neptune is; in a new one, the masses of the Sun and of the planets with satellites. (5) *No deductive entailment:* Newton's axioms do not entail K12 or K3, and with the planets' masses they contradict their exact form, as Duhem (1954) observed; with masses and initial conditions they entail honest tolerance forms, "K12 holds within $\varepsilon(b, W)$ on the window W" (**??** is a provable toy version). Formally deriving even one such bound needs an existence-and-uniqueness theorem and a validated integration or perturbation estimate, a very long derivation that no textbook writes out: a concrete case of the computational gap between a written theory and the reasoner who uses it (**??**).

A The Kepler computations

This appendix gives the proofs of the Kepler propositions, the definitions behind each computation, the full tables, and the elements that witness each reported fit. All computations take the plan94 analytical ephemeris of the planets and the moon98 ephemeris of the Moon, as implemented in the ERFA library, as the data; every positional error is against the plan94 ephemeris (moon98 for the Moon), not against the sky.

A.1 Proofs

The background B of Definition 1.1 cannot be the whole theory of the reals with sin, which is undecidable ($\sin(\pi x) = 0$ defines $\mathbb{Z}$); the numerical bounds it must prove are rational-function bounds of the kind MetiTarski proves (Akbarpour and Paulson, 2010).

The Ramsey sentence $R(T)$ of a theory T is obtained by replacing its theoretical function symbols by second-order variables and quantifying them existentially; its consequences are taken in second-order semantic consequence.

Proposition A.1 (The Ramsey sentence of Kepler's laws)**.**

(a) $R(T)$ has the same $\mathcal{L}_{\text{hel}}$-consequences as T.

(b) For every B true in $\mathbb{R}$, $R(\text{K12} \cup \text{K3}) \cup B$ entails no datum $\text{Near}(b, t, \mathbf{v}, \delta)$ with $\delta < \pi$.

Proof. (a) A model of $R(T)$ is an $\mathcal{L}_{\text{hel}}$-structure that expands to a model of T, so $R(T)$ and T have the same $\mathcal{L}_{\text{hel}}$-reducts of models. (b) Fix b, t, $\mathbf{v}$ and $\delta < \pi$, and a model of $B \cup \text{K12} \cup \text{K3}$ on the standard reals. If b is a planet, keep $a(b)$ and $P(b)$, so that K3 is untouched, and choose $i(b), \Omega(b), \varpi(b)$ and $M_0(b)$ so that the Keplerian direction of b at t is $-\mathbf{v}$: the orbital plane can be any plane through the Sun (the three angles give every rotation), and the phase can place the body at any point of its orbit, which meets every direction of its plane. If b is not a planet, $\text{K12} \cup \text{K3}$ does not constrain its direction at t (for the Sun it requires only $r(\text{Sun}, t) = 0$), and we set that direction to $-\mathbf{v}$. Defining θ, β, r from the positions gives a model of $B \cup \text{K12} \cup \text{K3}$, hence of $R(\text{K12} \cup \text{K3}) \cup B$, in which the direction of b at t makes the angle $\pi > \delta$ with $\mathbf{v}$. □

So the laws alone answer only structural questions; the laws together with the elements answer position questions.

Proof of Proposition 1.2. The position $\mathbf{r}(b,t)$ that K12 gives is $a(b)$ times an expression free of a, so a Kepler direction depends on $(e,i,\Omega,\varpi,M_0,P)$ and not on a. Given a model of $B\cup\mathrm{K12}$, put $a'(b) := (k^2P(b)^2/4\pi^2)^{1/3}$ for a fixed $k>0$ (cube roots exist in a real closed field) and leave every other element unchanged. Positions and distances scale by $a'(b)/a(b)>0$, directions are unchanged, and K3 holds because $a'(b)^3/P(b)^2 = k^2/4\pi^2$ is the same for every body. So an $\mathcal{L}_{\mathrm{dir}}$-sentence true in every model of $B\cup\mathrm{K12}\cup\mathrm{K3}$ is true in every model of $B\cup\mathrm{K12}$; the converse is trivial. □

Proposition A.2 (Content in geocentric directions)**.** *Let χ_{opp} be the sentence $\exists t\ \mathrm{dir}_g(\mathrm{Ma},t) \neq -\mathrm{dir}_g(\mathrm{Sun},t)$ of $\mathcal{L}_{\mathrm{geo}}$ ("Mars is not permanently in opposition").*

(a) χ_{opp} holds in every model of $\mathrm{K12}\cup\mathrm{K3}$ whose real sort is $\mathbb{R}$. Hence $B\cup\mathrm{K12}\cup\mathrm{K3}\vdash\chi_{\mathrm{opp}}$ for every B that contains the two true sentences ζ_{K3} and ζ_{pos} of the real field with sin *and* cos *given in the proof below.*

(b) For every B true in $\mathbb{R}$, $B\cup\mathrm{K12}\nvdash\chi_{\mathrm{opp}}$.

The sentence χ_{opp} is odd (its negation describes a fantastic world); it only shows that the geocentric language can register K3. The natural form of K3's content is Proposition 1.3.

Proof idea. (b) Circular coplanar orbits with equal periods and phases, Mars at distance $\rho>1$, satisfy K12 (which relates no bodies), violate K3 and keep Mars in opposition. (a) Over $\mathbb{R}$, a Mars permanently in opposition stays on the ray from the Sun through the Earth, and Kepler's second law forces $\mathbf{r}(\mathrm{Ma},t)=\kappa\,\mathbf{r}(\mathrm{Ea},t)$ with a constant $\kappa>1$: equal periods, different semi-major axes, against K3. ζ_{K3} states this about the elements, and ζ_{pos} (the Earth is never at the Sun) lets the argument cover non-standard models.

Proof. (b) Let $\mathcal{M}_0$ have the Earth with $a=1$, $e=0$, $i=0$, period P_0 and some phase, and Mars with $a=\rho>1$, $e=0$, $i=0$, the same period P_0 and the same phase. Then $\mathbf{r}(\mathrm{Ma},t)=\rho\,\mathbf{r}(\mathrm{Ea},t)$ for all t, so $\mathbf{r}(\mathrm{Ma},t)-\mathbf{r}(\mathrm{Ea},t)=(\rho-1)\mathbf{r}(\mathrm{Ea},t)\neq 0$ and $\mathrm{dir}_g(\mathrm{Ma},t)=\mathbf{r}(\mathrm{Ea},t)/|\mathbf{r}(\mathrm{Ea},t)| = -\mathrm{dir}_g(\mathrm{Sun},t)$ for every t. So $\mathcal{M}_0\models\neg\chi_{\mathrm{opp}}$. $\mathcal{M}_0$ satisfies K12, since each body moves on a Kepler orbit and K12 imposes no relation between bodies, and it violates K3 (equal periods, unequal semi-major axes). It is a model of every B true in $\mathbb{R}$, so $B\cup\mathrm{K12}\nvdash\chi_{\mathrm{opp}}$.

(a), standard models. Let $\mathcal{M}$ be a model of $\mathrm{K12}\cup\mathrm{K3}$ whose real sort is $\mathbb{R}$, and suppose $\mathcal{M}\models\neg\chi_{\mathrm{opp}}$. The Earth moves in the ecliptic ($i(\mathrm{Ea})=0$), so $\mathbf{u}(t) := \mathbf{r}(\mathrm{Ea},t)/|\mathbf{r}(\mathrm{Ea},t)|$ lies in the ecliptic and differs from $\mathbf{z}$. By $\neg\chi_{\mathrm{opp}}$, $\mathrm{dir}_g(\mathrm{Ma},t)=\mathbf{u}(t)\neq\mathbf{z}$ for all t, so the bridge gives $\mathbf{r}(\mathrm{Ma},t)\neq\mathbf{r}(\mathrm{Ea},t)$ and $\mathbf{r}(\mathrm{Ma},t)=\mathbf{r}(\mathrm{Ea},t)+c(t)\mathbf{u}(t)=(r_{\mathrm{E}}(t)+c(t))\,\mathbf{u}(t)$ with $c(t)>0$, where $r_b(t) := |\mathbf{r}(b,t)|$. So Mars is always on the ray from the Sun through the Earth, beyond the Earth: it moves in the ecliptic, its heliocentric direction equals the Earth's at every t, and $r_{\mathrm{M}}(t)>r_{\mathrm{E}}(t)$. Take continuous polar angles $\theta_{\mathrm{M}}=\theta_{\mathrm{E}}$ in the ecliptic. Kepler's second law, a consequence of K12, gives $r_b(t)^2\dot\theta_b(t)=h_b$ with a nonzero constant h_b for each body (from $r=a(1-e\cos\psi)$, $\tan\frac{\nu}{2}=\sqrt{(1+e)/(1-e)}\tan\frac{\psi}{2}$ for the true anomaly ν, and $\dot M=2\pi/P$, one gets $r^2\dot\nu=2\pi a^2\sqrt{1-e^2}/P$; standard). Hence $r_{\mathrm{M}}^2/r_{\mathrm{E}}^2=h_{\mathrm{M}}/h_{\mathrm{E}}$ is constant: $r_{\mathrm{M}}(t)=\kappa\,r_{\mathrm{E}}(t)$ with a constant $\kappa>1$, so $\mathbf{r}(\mathrm{Ma},t)=\kappa\,\mathbf{r}(\mathrm{Ea},t)$ for all t. The period of a Kepler orbit is the least period of $t\mapsto\mathbf{r}(b,t)$, and its semi-major axis is half the sum of its largest and smallest distances from the Sun ($r=a(1-e\cos\psi)$). So $P(\mathrm{Ma})=P(\mathrm{Ea})$ and $a(\mathrm{Ma})=\kappa\,a(\mathrm{Ea})\neq a(\mathrm{Ea})$, contradicting K3.

(a), arbitrary models. Let ζ_{K3} be the following sentence of the language of the real field with sin and cos:

$$\forall\vec\varepsilon_{\mathrm{Ea}}\,\forall\vec\varepsilon_{\mathrm{Ma}}\Big(\mathrm{Adm}(\vec\varepsilon_{\mathrm{Ea}},\vec\varepsilon_{\mathrm{Ma}})\to\exists t\,\forall\psi_{\mathrm{Ea}}\,\forall\psi_{\mathrm{Ma}}\big(\mathrm{Kep}_{\mathrm{Ea}}(t,\psi_{\mathrm{Ea}})\wedge\mathrm{Kep}_{\mathrm{Ma}}(t,\psi_{\mathrm{Ma}})\to\Delta(t,\psi_{\mathrm{Ea}},\psi_{\mathrm{Ma}})\big)\Big).$$

Here $\vec{\varepsilon}_b$ are the seven elements of body b; Adm says that $0 \le e < 1$ and $a, P > 0$ for both bodies, that $i_{\mathrm{Ea}} = 0$, and that K3 holds for the pair; $\mathrm{Kep}_b(t, \psi)$ is Kepler's equation $\psi - e_b \sin\psi = M_{0,b} + 2\pi t/P_b$; and Δ says that the positions $\boldsymbol{\xi}_b := \mathrm{Rot}_b(a_b(\cos\psi_b - e_b),\ a_b\sqrt{1-e_b^2}\sin\psi_b,\ 0)$ either coincide or satisfy $(\boldsymbol{\xi}_{\mathrm{Ma}} - \boldsymbol{\xi}_{\mathrm{Ea}})/|\boldsymbol{\xi}_{\mathrm{Ma}} - \boldsymbol{\xi}_{\mathrm{Ea}}| \neq \boldsymbol{\xi}_{\mathrm{Ea}}/|\boldsymbol{\xi}_{\mathrm{Ea}}|$. Let ζ_{pos} be the sentence $\forall a\, \forall e\, \forall \psi\, (a > 0 \wedge 0 \le e < 1 \to a(1 - e\cos\psi) > 0)$. Both are true in $\mathbb{R}$: ζ_{pos} obviously, and ζ_{K3} because, for standard element values satisfying Adm, Kepler's equation has exactly one solution for each t (its left side is strictly increasing in ψ when $0 \le e < 1$), so the antecedent of the inner implication fixes the true positions, and the standard case above gives a t at which they coincide or the directions differ.

Now let $\mathcal{M}$ be any model of $B \cup \mathrm{K12} \cup \mathrm{K3}$ with $\zeta_{\mathrm{K3}}, \zeta_{\mathrm{pos}} \in B$; its real sort may be non-standard. Instantiate ζ_{K3} at the elements of the Earth and Mars in $\mathcal{M}$ (Adm holds by K12, K3 and the frame convention) and obtain t. K12 supplies, for this t, values $\psi_{\mathrm{Ea}}, \psi_{\mathrm{Ma}}$ that solve Kepler's equations and give $\mathbf{r}(b, t) = \boldsymbol{\xi}_b$; the universal quantifiers over $\psi_{\mathrm{Ea}}, \psi_{\mathrm{Ma}}$ apply to them, so Δ holds of the actual positions. By ζ_{pos}, $|\mathbf{r}(\mathrm{Ea}, t)| = a_{\mathrm{Ea}}(1 - e_{\mathrm{Ea}}\cos\psi_{\mathrm{Ea}}) > 0$, so $-\mathrm{dir}_g(\mathrm{Sun}, t) = \mathbf{r}(\mathrm{Ea}, t)/|\mathbf{r}(\mathrm{Ea}, t)|$. If the positions coincide, the bridge gives $\mathrm{dir}_g(\mathrm{Ma}, t) = \mathbf{z}$, which differs from $\mathbf{r}(\mathrm{Ea}, t)/|\mathbf{r}(\mathrm{Ea}, t)|$, whose third coordinate is 0 since $i_{\mathrm{Ea}} = 0$. Otherwise the bridge gives $\mathrm{dir}_g(\mathrm{Ma}, t) \neq \mathbf{r}(\mathrm{Ea}, t)/|\mathbf{r}(\mathrm{Ea}, t)|$. Either way $\mathcal{M} \models \chi_{\mathrm{opp}}$, and by the completeness theorem $B \cup \mathrm{K12} \cup \mathrm{K3} \vdash \chi_{\mathrm{opp}}$. □

The universal quantifier over ψ in ζ_{K3} is what makes the instantiation valid in a model in which Kepler's equation might have several solutions; an existential quantifier would leave that case open. The sentence ζ_{pos} supplies $\mathbf{r}(\mathrm{Ea}, t) \neq 0$ in non-standard models, the only fact about the reals used outside ζ_{K3}.

Remarks on Proposition A.2. (i) The bridge needs a convention at coincidence: without it, a K3-model in which Mars coincides with the Earth would leave $\mathrm{dir}_g(\mathrm{Ma}, t)$ unconstrained there and could imitate $\mathcal{M}_0$. (ii) For an inner co-moving planet ($\rho < 1$) the analogue fails: a K3-model with the planet on the Earth's orbit, diametrically opposite (behind the Sun), is always in the direction of the Sun; so χ_{opp} is stated for an outer planet. (iii) A computation (code in the accompanying repository) illustrates the difference in the circular coplanar case with $\rho = 1.5237$: $\mathcal{M}_0$ keeps Mars at opposition to rounding error (1.2×10^{-6} degrees in floating point), while the K3-model with the same Earth, the same ρ and the same initial phase leaves opposition by 1.3° within a day, 13° within 10 days and 87° within 100 days. A second, independently written implementation finds, over 300 random $\mathrm{K12} \cup \mathrm{K3}$ models (an outer planet, the Earth in the ecliptic, the planet's plane random), a largest angle from opposition over 20 years of at least 175° in every model, and over 50 random models of the kind of $\mathcal{M}_0$ never more than 6×10^{-14} degrees. Whether finitely many interval data true in $\mathcal{M}_0$ refute every K3-model was not examined.

Proof of Proposition 1.3. (a) K3 with $a(\mathrm{Ea}) = 1$ gives $a(\mathrm{Ma})^3 = P(\mathrm{Ma})^2/P(\mathrm{Ea})^2$, so $\rho = (P(\mathrm{Ma})/P(\mathrm{Ea}))^{2/3}$; then K12 and Num^- determine $\mathbf{r}(\mathrm{Ma}, t)$ and $\mathbf{r}(\mathrm{Ea}, t)$, and the bridge determines the geocentric direction. A datum $\mathrm{Near}_g(\mathrm{Ma}, t, \mathbf{v}, \delta)$ satisfied by the computed direction then follows once B proves the numerical bound that places the computed direction within δ of $\mathbf{v}$.

(b) Fix t with $\mathbf{r}(\mathrm{Ea}, t)$ and $\mathbf{F}(t)$ not parallel. Every $\rho > 0$ gives a model of $\mathrm{K12} \cup \mathrm{Num}^-$ over $\mathbb{R}$, since K12 constrains a only by $a > 0$. The geocentric vector is $\mathbf{g}(\rho) = \rho\mathbf{F}(t) - \mathbf{r}(\mathrm{Ea}, t)$; as ρ runs over $(0, \infty)$ it runs along the open ray from $-\mathbf{r}(\mathrm{Ea}, t)$ in the direction $\mathbf{F}(t)$, inside the plane spanned by the two vectors. The ray does not pass through the origin because the two vectors are not parallel. Seen from the origin, its direction therefore moves continuously and

monotonically along the great circle through $-\mathbf{u}(t)$ and $\mathbf{F}(t)/|\mathbf{F}(t)|$, tends to $-\mathbf{u}(t)$ as $\rho \to 0$ and to $\mathbf{F}(t)/|\mathbf{F}(t)|$ as $\rho \to \infty$, and covers the open arc between them, whose length $\alpha(t)$ lies in $(0, \pi)$. If $B \cup \mathrm{K12} \cup \mathrm{Num}^-$ proved $\mathrm{Near}_g(\mathrm{Ma}, t, \mathbf{v}, \delta)$ for a B true in $\mathbb{R}$, the whole arc would lie in the closed cap of angular radius δ about $\mathbf{v}$; its two end points, at angular distance $\alpha(t)$ from each other, would then lie in that cap too, so $\alpha(t) \le 2\delta$. □

A computation over 1580–1600 (daily, plan94) gives the distribution of $\alpha(t)$ quoted in Example A.4: percentiles $0, 5, 25, 50, 75, 95, 100$ of $0.11°$, $9.52°$, $46.68°$, $90.22°$, $133.91°$, $170.34°$, $179.96°$; no sampled day is within $2'$ of conjunction or opposition. Along the arc the direction moves monotonically as ρ grows on every one of 21 sampled days (a second implementation finds the same on 60 seeded days, with the great-circle identity to 9×10^{-15} radians), and the sensitivity of the geocentric direction to ρ at the true ρ has median $969'$ and maximum $3555'$ per unit of ρ.

A.2 Data, fits and the error budget

Table 2: How wrong Kepler's laws are, against plan94: largest angular separation (arcminutes) between one fixed Kepler ellipse, fitted by least squares, and the plan94 positions, over windows centred on J2000; for the Moon (moon98, geocentric), in degrees. Last column: the fitted third-law constant $q := (2\pi/P)^2 a^3/k^2$ (k Gauss's constant) over the windows of 30 years and more, with the Newtonian two-body value $1 + m/M$ in brackets. Least-squares fits, angular separation; the minimax longitude errors of the same windows, used in **??**, are in **??**.

window (years)	12	30	100	300	1000	2000	third-law constant q
Mercury	0.46	0.89	2.17	6.04	19.3	38.6	
Mars	1.58	1.86	3.61	9.17	26.2	51.8	
Jupiter	1.05	4.19	7.64	7.74	31.8	38.5	1.00075–1.00099 (1.00095)
Saturn	1.70	3.32	9.73	11.7	91.3	92.4	1.0013–1.0019 (1.00029)
Moon	1.12° (one month), 3.0° (one year), 9.6° (18.6 years)						

Residuals of one fixed ellipse. For each body and window, centred on J2000, one fixed Kepler ellipse (seven elements) is fitted by least squares to the plan94 heliocentric positions (for the Moon, the moon98 geocentric positions) in the J2000 ecliptic, and the table reports the largest angular separation between the fit and the data. The fitted third-law constant is $q := (2\pi/P)^2 a^3/k^2$, with k Gauss's constant. Over the 12-year window the fits give $q = 1.00062$ (Jupiter) and 1.00487 (Saturn); over the windows of 30 years and more they give the ranges of the table. The Newtonian two-body value is $1 + m/M$; for Saturn, a secular, circular and coplanar estimate that treats the orbit-averaged pull of each other planet as that of a ring gives $q = 1.00154$, of which Jupiter supplies 1.25×10^{-3}. This is a heuristic estimate, not a perturbation theory. A second computation, with other sample times and longitude errors only, agrees with the table to within about 8% in every planetary cell. The same fits, with a precessing ellipse, give Mercury's line of apsides a rotation of $575.4''$ per century in the fixed J2000 frame (fits over 100–2000 years), and the 2000-year longitude residuals of Saturn and Jupiter carry oscillations of period about 900 and 910 years and amplitude $43.8'$ and $18.3'$.

The third law's own error. With $q - 1 = 9.5 \times 10^{-4}$ (Jupiter, the two-body value) and 1.5×10^{-3} (Saturn, the middle of its fitted range), an exact K3 calibrated on the Earth changes

the semi-major axes by the relative amounts 3.170×10^{-4} and 4.985×10^{-4} and misplaces Jupiter and Saturn geocentrically by at most $0.233'$ and $0.193'$ over 1580–1600 (medians $0.150'$ and $0.129'$).

The error budget. The figures of this paragraph are those quoted in the documentation of the ERFA routine `eraPlan94` (which pyerfa wraps): from comparisons with JPL's DE102 ephemeris over 1800–2050, maximum errors in heliocentric longitude of $17''$ (Mars), $71''$ (Jupiter) and $81''$ (Saturn); an accuracy over 1000–3000 no worse than 1.5 times that, and, against DE200 over 1800–2100, $26''$, $78''$ and $87''$. Hence: (i) over 1580–1600 plan94's longitude error for Mars is at most about $25'' \approx 0.43'$, the size of the gap $1.41' - 1.07' = 0.34'$ between the two best forms of the Mars pool, so the ordering of those two forms is established against plan94, and against the sky it rests on the errors against an exact ellipse in Table 3; (ii) the 12-year entries for Jupiter ($1.05'$) and Saturn ($1.70'$) lie inside 1800–2050, where plan94's own errors reach $1.18'$ and $1.35'$, so they measure the ellipse against plan94; (iii) plan94's Mars distance error (at most 7700 km over 1800–2050, about 3×10^{-5} of Mars's distance) is far below the distance tolerances of Appendix A.6.

The data of the Mars pool. The 7306 daily heliocentric positions of Mars over 1580.0–1600.0 are reduced to the best plane through the Sun (inclination 1.8828° and node 50.757° to the J2000 ecliptic; the largest angle of a position out of this plane is $0.070'$), and v_i is the longitude in that plane. The geocentric data of Appendix A.5 are the ecliptic longitudes of plan94 Mars minus plan94 Earth–Moon barycentre (J2000 ecliptic): geometric positions, without aberration or light time, and taken from the barycentre rather than from the Earth.

A.3 The Mars pool: forms, law texts and Occam factors

The forms. Write $\mathbf{u}(\varphi) := (\cos\varphi, \sin\varphi)$ and $\Lambda(t) := L + n(t - t_{\mathrm{mid}})$ for the mean longitude, with t_{mid} the middle of the window, L the mean longitude at t_{mid} and n the mean motion; w is the longitude of perihelion and $M(t) := \Lambda(t) - w$ the mean anomaly. Each form gives the heliocentric longitude $\theta(t)$ in the plane of the data; in longitude the forms are scale-free, so the circle has radius 1. The forms are heliocentric reconstructions: the historical models before Kepler were geocentric and also represented the Earth's motion.

- *Circle about the Sun* (elements L, n): $\theta(t) = \Lambda(t)$.
- *Circle with centre* $C = -e_1\mathbf{u}(w)$ *and equant point* $Q = -(e_1 + e_2)\mathbf{u}(w)$, both on the line of apsides on the aphelion side of the Sun: the planet moves on the unit circle about C so that its direction seen from Q is $\Lambda(t)$, and $\theta(t)$ is its direction seen from the Sun. The *eccentric circle* is the case $e_2 = 0$ (uniform motion about the centre; elements e, w, L, n with $e = e_1$), the *bisected equant* the case $e_1 = e_2 = e$ (centre halfway between the Sun and the equant point; elements e, w, L, n), and the *vicarious hypothesis* the case of free e_1, e_2 (elements e_1, e_2, w, L, n).
- *Ellipse* (elements e, w, L, n): $\theta(t) = w + \nu$, where ν is the true anomaly that Kepler's equation $\psi - e\sin\psi = M(t)$ gives for eccentricity e.

The element ranges are $e, e_1, e_2 \in [0, 1)$, $w, L \in [0, 2\pi)$ and $n \in [0, 0.1)$ radians per day. Each element is written as a fixed-length 64-bit binary numeral for the midpoint of a cell of its range. The weight of the compatible numerals of a form is then $\mathrm{Vol}_{\mathsf{F}}(\delta)/\prod(\text{ranges})$ up to a grid error of order 2^{-64}, which gives the posterior weight $2^{-(|\mathrm{LAW}_{\mathsf{F}}|_S + \mathrm{Occ}_{\mathsf{F}}(\delta))}$ of Section 1.4.

Minimax errors. Each form is fitted by least squares from several starts, refined by L_p fits with increasing p, and its largest residual is then minimised by sequential linear programming.

Table 3 lists the minimax errors and the elements that witness them; each value against plan94 is an upper bound on the true minimax error. A second implementation, an exchange algorithm written independently and started elsewhere, finds the same values to four decimals. The circle's value is exact: its minimax problem is a one-dimensional convex minimisation. For the others, global optimality is not proved. The column "exact" gives the minimax errors against an exact, unperturbed Kepler ellipse (the ellipse's own least-squares elements on these data, $e = 0.0930$, whose largest deviation from plan94 is $1.25'$); so more than half of the vicarious hypothesis's $1.41'$ is perturbation error shared with the ellipse. The eccentric's error against the exact ellipse agrees with the second-order expansions of the two equations of centre, $2e^2 \sin 2M$ for the eccentric and $\frac{5}{4}e^2 \sin 2M$ for the ellipse, whose difference $\frac{3}{4}e^2$ is about $22.5'$ for Mars.

Table 3: Minimax errors of the Mars pool, 1580–1600 (arcminutes): against plan94, upper bounds witnessed by the elements shown (w and L in degrees, L at the middle of the window; n in radians per day), and against an exact ellipse; k is the number of elements.

form	k	plan94	exact	witnessing elements
circle	2	640.7908	–	$L = 309.45$, $n = 0.00914611$
eccentric	4	23.0508	22.34	$e = 0.184516$, $w = 283.471$, $L = 309.448$, $n = 0.0091461$
equant	4	8.1653	7.43	$e = 0.0933767$, $w = 283.465$, $L = 309.446$, $n = 0.00914613$
vicarious	5	1.4127	0.62	$e_1 = 0.115724$, $e_2 = 0.0707907$, $w = 283.462$, $L = 309.446$, $n = 0.00914609$
ellipse	4	1.0733	0	$e = 0.0930389$, $w = 283.445$, $L = 309.442$, $n = 0.00914609$

The law texts. A law text is a sentence $\forall b \, \forall t \, (\theta(b,t) = \ldots)$, or with an existential quantifier as below, in which the elements are unary function symbols $L(b), n(b), w(b), e(b), e_1(b), e_2(b)$ and the mean longitude is written out as $L(b) + n(b) \cdot t$, with t measured from the middle of the window, wherever it occurs (below we abbreviate it by Λ, and $\Lambda - w(b)$ by M; the abbreviations are expanded in the counted texts). The syntaxes are: *S1*, with $+, -, \cdot, /, \sin, \cos, \mathrm{atan2}, \sqrt{\ }$, the numerals $0, 1, 2$, $\forall$, $\exists$, $\wedge$, $=$, $>$; *S2*, S1 with a unit-vector primitive $\mathrm{uc}(\varphi) = \mathbf{u}(\varphi)$, a direction primitive $\mathrm{dir}(\mathbf{x})$ (the polar angle of $\mathbf{x}$) and $\mathrm{rc}(Q, C, \varphi)$, the point where the ray from Q at angle φ meets the unit circle about C; and *S3*, S1 with $\mathrm{kep}(e, M)$, the true anomaly for eccentricity e and mean anomaly M. The texts are (arguments b of the element symbols omitted):

- circle (all syntaxes): $\theta(b,t) = L + n \cdot t$;
- eccentric: in S1 and S3, $\theta(b,t) = \mathrm{atan2}(\sin\Lambda - e \cdot \sin w, \ \cos\Lambda - e \cdot \cos w)$; in S2, $\theta(b,t) = \mathrm{dir}(\mathrm{uc}(\Lambda) + (-e) \cdot \mathrm{uc}(w))$;
- vicarious hypothesis: in S1 and S3, $\exists s \, (s > 0 \wedge s \cdot s - 2 \cdot (e_2 \cdot (s \cdot \cos M)) + e_2 \cdot e_2 = 1 \wedge \theta(b,t) = \mathrm{atan2}(-((e_1+e_2) \cdot \sin w) + s \cdot \sin\Lambda, \ -((e_1+e_2) \cdot \cos w) + s \cdot \cos\Lambda))$; in S2, $\theta(b,t) = \mathrm{dir}(\mathrm{rc}(-(e_1 + e_2) \cdot \mathrm{uc}(w), \ (-e_1) \cdot \mathrm{uc}(w), \ \Lambda))$;
- bisected equant: the vicarious texts with e for e_1 and e_2 in S1 and S3, and with $2 \cdot e$ for $e_1 + e_2$ and e for e_1 in S2;
- ellipse: in S1 and S2, $\exists E \, (E - e \cdot \sin E = M \wedge \theta(b,t) = w + \mathrm{atan2}(\sqrt{1 - e \cdot e} \cdot \sin E, \ \cos E - e))$; in S3, $\theta(b,t) = w + \mathrm{kep}(e, M)$;
- K3 (all syntaxes; with $P = 2\pi/n$): $\forall b \, \forall c \, ((n(b) \cdot n(b)) \cdot (a(b) \cdot (a(b) \cdot a(b))) = (n(c) \cdot n(c)) \cdot (a(c) \cdot (a(c) \cdot a(c))))$.

A text is serialised with one opening and one closing bracket per compound, and its length is its number of tokens times $\lceil \log_2(\text{alphabet size}) \rceil$. The alphabets have 31 (S1), 34 (S2) and 32 (S3) symbols, so a token costs 5, 6 and 5 bits. This gives the text lengths of Table 1: 31, 89, 198, 198, 109 and 75 tokens in S1 for the circle, eccentric, equant, vicarious hypothesis, ellipse and K3; 31, 57, 75, 78, 109, 75 in S2; 31, 89, 198, 198, 52, 75 in S3. Every text was evaluated numerically and compared with the model it is attached to, at 40 random elements and times each: all agree to within 1.4×10^{-14} radians. A second serialisation, Polish notation with fixed arities and a separate symbol for unary minus, gives about half these lengths: in S1 75, 195, 440, 440, 245 and 165 bits, in S2 90, 150, 198, 204, 294, 198 and in S3 75, 195, 440, 440, 120, 165; a compact Polish code for K3 with a power symbol and the numeral 3 has 23 tokens, 115 bits.

Occam factors. $\mathrm{Occ}_{\mathsf{F}}(\delta)$ is estimated by hit-or-miss Monte Carlo with the exact model on all 7306 data: points are drawn uniformly in a parallelepiped built from the whitened linearisation of the residuals at the minimax point, enlarged where hits come close to a face, and capped at the element ranges; about 2000 hits per entry give a standard error of about 0.03 bits. The table below gives every value. Two cross-checks support them where the region is far from linear. Hit-or-miss sampling in an axis-aligned box with the full range of the eccentricities gives, for the vicarious hypothesis, 22.73 ± 0.03, 33.36 ± 0.03, 37.87 ± 0.07 and 43.02 ± 0.24 bits at $720'$, $120'$, $60'$ and $30'$, within about one combined standard error of the table. An independent implementation, with boxes found by its own random-walk explorer, gives 21.16 (ellipse, $720'$), 20.21 (eccentric, $720'$), 21.17 (equant, $720'$), 31.72 (equant, $120'$), 31.69 (ellipse, $120'$), 35.67 (ellipse, $60'$), each ± 0.03 or better, and 20.929 (circle, $720'$, an exact one-dimensional integral): all within 0.06 bits of the table. At $720'$ the compatible regions have a thin part with the apsidal angle opposite the true perihelion and small eccentricity (since the circle fits to $640.8' < 720'$, every apsidal angle is compatible for small e); the parallelepiped at $720'$ spans 6.5 radians in w, and a full-range rerun shows the effect of this part to be within the noise.

δ	$720'$	$120'$	$60'$	$30'$	$20'$	$12'$	$9'$	$6'$	$4'$	$3'$	$2'$	$1.6'$	$1.3'$
circle	20.93	–	–	–	–	–	–	–	–	–	–	–	–
eccentric	20.24	31.17	36.34	45.38	–	–	–	–	–	–	–	–	–
equant	21.20	31.70	35.88	40.35	43.63	49.49	57.34	–	–	–	–	–	–
vicarious	22.77	33.34	37.93	42.83	45.77	49.56	51.78	54.98	58.56	61.53	67.56	74.54	–
ellipse	21.18	31.63	35.68	39.65	42.01	45.01	46.81	49.21	51.86	53.88	57.41	59.93	63.81

Occam factors $\mathrm{Occ}_{\mathsf{F}}(\delta)$ in bits ("–": δ below the form's minimax error).

The MAP form. At these 13 precisions the MAP form is: in S1, the circle at $720'$, the eccentric at $120'$–$30'$ and the ellipse from $20'$ on; in S2, the circle at $720'$, the eccentric at $120'$–$30'$, the bisected equant at $20'$–$9'$, the vicarious hypothesis at $6'$–$1.6'$ and the ellipse at $1.3'$; in S3, the circle at $720'$ and the ellipse from $120'$ on. Below $1.07'$ no form is compatible. In every row the runner-up has at most $2^{-12.4}$ of the MAP's weight. Each K3 variant of a form has exactly $2^{-|\mathrm{K3}|}$ times the form's weight: with one planet K3 is vacuous, and in heliocentric direction data it has no content (Proposition 1.2).

The data without the text prior. With a uniform prior over the five forms, the element prior above and no text prior, the MAP is the eccentric at $720'$ and $120'$ (margins 0.69 and 0.46 bits) and the ellipse at every $\delta \le 60'$, with margins 0.20, 0.70, 1.62, 4.48, 4.97, 5.77, 6.70, 7.65, 10.15 and 14.61 bits at $60'$, $30'$, $20'$, $12'$, $9'$, $6'$, $4'$, $3'$, $2'$ and $1.6'$. These margins are largest just

above a competitor's minimax error, where its compatible volume shrinks to zero. They depend on the element ranges: halving the ranges of the eccentricities (the compatible regions lie inside $[0, 0.5)$ at every $\delta \leq 120'$) adds one bit of prior density per eccentricity, which moves the vicarious hypothesis by one bit against the four-element forms and the circle by one bit the other way.

Text against data. Form F beats form G iff $|\mathrm{LAW}_{\mathsf{F}}|_S - |\mathrm{LAW}_{\mathsf{G}}|_S < \mathrm{Occ}_{\mathsf{G}}(\delta) - \mathrm{Occ}_{\mathsf{F}}(\delta)$. For the epicyclic syntax's equant and vicarious texts (18 bits apart), the Occam-factor difference moves from 2.14 bits in the equant's favour at $20'$ to 5.56 bits in the vicarious hypothesis's favour at $9'$, so the equant keeps a margin of 12.4 bits at $9'$; text and data contributions are of the same order there. For every other pair of forms the text difference is between 100 and 835 bits, an order of magnitude larger than any Occam-factor difference, except that in S1 and S3 the equant and vicarious texts have equal length; both lose to the ellipse there by more than 400 bits.

Robustness. A variable-length numeral code (an Elias-gamma length prefix per element) adds about 5–11 bits per element; this moves the vicarious hypothesis against the four-element forms by 6–11 bits and the circle against them by 12–20 bits, and changes none of the MAP rows. In Polish notation the text differences shrink by about half and no MAP row changes; the closest case is the epicyclic syntax's equant and vicarious texts, which then differ by 6 bits (198 against 204), against an Occam-factor difference of at most 5.56 bits. A natural form outside the pool, the equation-of-centre series truncated at second or third order (four elements, like the ellipse), has S1 texts of 680 and 1345 bits, against the ellipse's 545, and minimax errors $4.11'$ and $1.31'$, so it changes no S1 row.

A.4 Graded fit

Example A.3 (Graded fit; a computation)**.** Under dithered rounding every form in the Mars pool of Section 1.4, the ellipse included, is eventually refuted (**??**); but inside M_A a theory may be a sampler, the law with a Gaussian error model of stated σ (**??**). At $\sigma = 2'$, roughly Tycho's precision, the expected penalty per independent observation relative to the ellipse is 45.05 bits (eccentric), 4.96 (equant) and 0.035 (vicarious hypothesis), so the text advantages over the ellipse are overcome after 2 and 7 observations (eccentric, S1 and S2), 41 (equant, S2) and about 5,300 (vicarious hypothesis, S2); a simulation agrees. Honest uniform-tolerance theories give 23 to 71 and about 470 data, a conservative estimate, since SAI's mixture prefers a law with a calibrated error model (**??**; details below). So with graded fit the data, not the syntax, decide every step within tens of observations, except the epicyclic syntax's choice between the vicarious hypothesis and the ellipse on longitudes, which needs thousands of longitudes at Tycho-level noise.

Gaussian noise. Least-squares fits to the 7306 longitudes have root-mean-square residuals $454.4'$ (circle), $15.81'$ (eccentric), $5.26'$ (equant), $0.628'$ (vicarious hypothesis) and $0.446'$ (ellipse). For a sampler with Gaussian errors of standard deviation σ, the expected log-likelihood penalty per independent observation relative to the ellipse is $(\mathrm{RMS}^2_{\mathsf{F}} - \mathrm{RMS}^2_{\mathrm{ell}})/(2\sigma^2 \ln 2)$ bits. A form's text advantage over the ellipse is overcome after (text advantage)/(penalty) observations, which omits the parameter-fitting terms:

form	penalty (bits per observation) $\sigma = 1'$	$\sigma = 2'$	$\sigma = 4'$	observations needed to overcome the text advantage at $\sigma = 1', 2', 4'$
eccentric	180.2	45.05	11.26	S1 (100 bits): 0.6, 2.2, 8.9; S2 (312 bits): 1.7, 6.9, 27.7
equant	19.84	4.96	1.24	S2 (204 bits): 10.3, 41.1, 164.5
vicarious	0.142	0.035	0.0089	S2 (186 bits): 1313, 5253, 21013

In S1 and S3 the ellipse already has the shorter text against the equant and the vicarious hypothesis, by 445 and 730 bits, and in S3 also against the eccentric.

A simulation. For n days drawn at random from 1580–1600 (seeded; 20 runs per cell), the plan94 longitude plus Gaussian noise is fitted by each form by least squares, and the evidence is computed by the Laplace approximation under the element prior above; the noise model's text is common to all forms and cancels. At $\sigma = 2'$, in S1 and S3 the MAP is the ellipse in 20 of 20 runs for every $n \geq 10$ (at $n = 5$, S1's MAP is the eccentric in all runs); in S2 it is the vicarious hypothesis in 20 of 20 runs for $n = 20$ to 2560, in 10 of 20 at $n = 5120$, and the ellipse in 20 of 20 at $n = 7306$. At $\sigma = 1'$ the S2 switch to the ellipse comes between $n = 640$ (vicarious hypothesis in 20 of 20) and $n = 2560$ (ellipse in 20 of 20). The median evidence ratio of the ellipse over the vicarious hypothesis, without text prior, grows from $2^{2.2}$ at $n = 5$ to 2^{276} at $n = 7306$ ($\sigma = 2'$). The simulation agrees with the expected-penalty estimates.

Honest uniform-tolerance theories. Give each form the honest law with tolerance equal to its minimax error (**??**). For cells much narrower than the tolerances, the cost per datum against the ellipse is about $\log_2(\varepsilon_{\mathsf{F}}/\varepsilon_{\text{ell}})$: 4.43 bits (eccentric), 2.93 (equant) and 0.40 (vicarious hypothesis). The text advantages are then overtaken after 23 (S1) and 71 (S2) data for the eccentric, 70 for the equant (S2) and about 470 for the vicarious hypothesis (S2). These are estimates from the pricing of **??**, not a separate posterior computation, and they are conservative: SAI's mixture prefers a law with a calibrated error model to the uniform band (**??**).

A.5 What K3 is worth

Example A.4 (K3's worth against its cost; a computation)**.** By Proposition 1.3, given the other elements, K3's geocentric content is exactly ρ, and it is large: over 1580–1600 the median of $\alpha(t)$ is 90° (5th and 95th percentiles 9.5° and 170°). Take as data the daily geocentric longitudes of Mars over 1580.0–1600.0 at a common tolerance δ, and Kepler ellipses for the Earth and Mars with $\rho \in [0, 4)$ free or given by K3. With fixed-length numerals for the elements, the posterior odds for K3 are $2^{s_{\text{K3}}(\delta) - |\text{K3}|}$, where $|\text{K3}|$ is the length of its text and $s_{\text{K3}}(\delta) := \log(4\,\text{Vol}_8(\delta)/\text{Vol}_9(\delta))$ ($\text{Vol}_8, \text{Vol}_9$ the volumes of the compatible element regions with and without K3) is what K3 saves (the method is described below). A computation gives $s_{\text{K3}} = 11.3$, 12.3, 13.4, 14.7 bits at $\delta = 8'$, $4'$, $2'$, $1'$, about one more bit per halving of δ, as for any real-valued element; K3 costs $0.005'$ of fit. Its text costs 375 bits in the syntax S1 of Section 1.4 (brackets counted), 165 in Polish notation and 115 in a compact Polish code; it is paid once and saves $s_{\text{K3}}(\delta)$ bits per body whose size it fixes. By **??** (with the law in place, at exchange rate 1; a comparison of law choices, not membership of a steeper MAP), K3 pays once it fixes the sizes of about 25–33, 11–15 or 8–10 bodies in the three codes, while the five planets other than the Earth give about 55–75 bits if each saves about as much as Mars (an extrapolation; we computed Mars only). So in none of these codes does compression alone select K3 at Kepler's precision. This is conditional on the code: under the universal prior the relevant cost is the complexity of K3 given the rest of the theory, which these computations do not determine.

Setting. The data are those of Example A.4 (Appendix A.2). The hypotheses are Kepler ellipses for the Earth (in the ecliptic, $a = 1$) and for Mars (its plane fixed at the best heliocentric plane of Appendix A.2), each with elements e, w, L, n, and either $\rho = a(\mathrm{Ma})/a(\mathrm{Ea})$ as a ninth free element in $[0, 4)$ (without K3) or $\rho := (n_{\mathrm{Ea}}/n_{\mathrm{Ma}})^{2/3}$ (with K3, eight elements). As in Appendix A.3, the numerals are of fixed length, which gives the posterior odds $2^{s_{\mathrm{K3}}(\delta)-|\mathrm{K3}|}$ of Example A.4, with Vol_8 and Vol_9 the volumes of the exact compatible regions with and without K3.

Fit. The minimax geocentric error is $0.627'$ with ρ free (fitted $\rho = 1.523676$, against the K3 value 1.523687 from the fitted periods) and $0.632'$ with K3; an independent fit gives $0.6266'$ and $0.6318'$ with the same values of ρ. So K3 costs $0.005'$ of fit and is compatible with the data at every $\delta \geq 0.632'$. These minimax values are upper bounds witnessed by elements. The data are values of a model, so plan94's own error does not affect the comparison.

Method. With $y := \rho - (n_{\mathrm{Ea}}/n_{\mathrm{Ma}})^{2/3}$, a shear of unit Jacobian turns the volume ratio into a density: $\mathrm{Vol}_8/\mathrm{Vol}_9 = p_y(0)$, the density at 0 of y under the uniform distribution on the nine-dimensional compatible region. That distribution is sampled by hit-and-run (Smith, 1984), with stepping-out brackets in the manner of slice sampling (Neal, 2003) and the exact membership test on all data; $p_y(0)$ is estimated as the fraction of samples with $|y| < h$ divided by $2h$, for $h = 0.05$, 0.1 and 0.2 standard deviations of y. Box sampling cannot be used here: the linearised compatible region fills a fraction of 10^{-5} or less of its bounding box.

δ	$8'$	$4'$	$2'$	$1'$
$s_{\mathrm{K3}}(\delta)$ (bits)	11.28 ± 0.04	12.31 ± 0.03	13.39 ± 0.03	14.73 ± 0.04
second implementation	11.27–11.32	12.28–12.32	13.32–13.35	14.69–14.71
standard deviation of y	$6.47 \cdot 10^{-4}$	$3.10 \cdot 10^{-4}$	$1.47 \cdot 10^{-4}$	$5.54 \cdot 10^{-5}$
$\log_2\left(4/(\sqrt{2\pi}\,\mathrm{sd}(y))\right)$	11.27	12.33	13.41	14.81

The fourth row is the value for a Gaussian y, as a check. The errors in the first row are one standard error over the 492–583 independent chains (of 600) that start at exactly compatible points. The three kernel widths agree to within 0.03 bits, and the first and second halves of the kept steps agree to within 0.06 bits except at $8'$ (11.33 against 11.23), where we quote 11.3 ± 0.1. The second implementation, written independently and run with three kernel widths, is a hit-and-run on the linearised compatible polytope with exact chords (60,000 samples per δ); its shares of linearised samples that are exactly compatible, 0.819, 0.901, 0.958, 0.968, match the first implementation's 0.820, 0.895, 0.948, 0.972. The sampler failed to bracket or to accept in 38, 5, 1 and 0 of about 128,000–152,000 chain steps per row; the direct effect is at most about 0.01 bits at $8'$. A cruder estimate from the single most sensitive day (9.8, 10.8, 11.8 bits at $8'$, $4'$, $2'$) is lower, as it should be. s_{K3} includes $\log_2$ of the range of ρ ($[0, 4)$, 2 bits); the range $[0, 2)$ would lower every entry by one bit.

Against the text. |K3| is 375 bits in S1 with brackets counted (450 in S2), 165 bits in Polish notation and 115 bits in the compact Polish code (Appendix A.3). For one pair of planets the posterior odds against K3 at $8'$ are therefore about 2^{-364} (S1) to 2^{-104} (compact code). By the criterion of **??**(b), when the same bodies' elements are in the theory with and without K3, K3 pays, with the law in place, iff the number $|S|$ of bodies whose size it fixes satisfies $s_{\mathrm{K3}}|S| > |\mathrm{K3}|$; at $s_{\mathrm{K3}} = 11.3$ and 14.7 bits per body this needs $|S| > 33.2$ and 25.5 (375 bits), 14.6 and 11.2 (165 bits), 10.2 and 7.8 (115 bits). The extrapolation to the five planets other than the Earth

(about 55–75 bits) assumes that each saves about as much as Mars; the other planets were not computed.

A.6 Distances as data

The data are those of the Mars pool together with, for every day, the sentence "$r(t_i)$ lies within a factor $1 \pm \tau_{\mathrm{d}}$ of r_i", with r_i the plan94 heliocentric distance in the best plane. Each form gets one more element, its scale. An entry of Table 4 is the least longitude tolerance found at which a form, with suitable elements and scale, proves every longitude datum and every distance datum. The search combines sequential linear programming from many starts: earlier optima, perturbations of them, the optimum at the neighbouring τ_{d} in a decreasing and in an increasing sweep, and six seeded random starts; a second, independently written search was run as well, and each entry is the best value found by either. Every entry is witnessed by printed elements and was re-checked with the same residual function; none is proved minimal. The circle about the Sun is not in this computation. The equant and ellipse rows come from a coarser search; their entries at 3%, 1.5% and 0.75% follow by monotonicity (elements feasible at a smaller τ_{d} are feasible at a larger one, and no form beats its longitude minimax), and the independent search confirms them: at its longitude minimax the ellipse needs only a distance tolerance of 0.0153% and the equant 0.2487%.

Table 4: Distances as data: the least longitude tolerance found (arcminutes, against plan94) at which a form proves every longitude datum and every distance datum at relative distance tolerance τ_{d}; "none": no elements meeting τ_{d} were found; r^*: the least distance tolerance found with no longitude constraint. Every entry is an upper bound on the least tolerance, witnessed by elements; minimality is not proved.

form	r^*	5%	3%	2%	1.5%	1%	0.75%	0.5%	0.2%	0.1%	0.05%
eccentric	0.226%	150.73	218.31	252.45	269.56	286.70	295.32	305.33	none	none	none
equant	0.222%	8.17	8.17	8.17	8.17	8.17	8.17	8.17	none	none	none
vicarious	0.0164%	1.41	1.41	2.01	3.54	5.12	5.97	6.95	23.74	94.95	132.04
ellipse	0.0117%	1.07	1.07	1.07	1.07	1.07	1.07	1.07	1.07	1.07	1.07

What the distances decide. At $\tau_{\mathrm{d}} = 1\%$ the least longitude tolerance at which the vicarious hypothesis fits all the data is at most $5.12'$ (witnessed by elements; two independent searches found nothing lower), while the ellipse fits both kinds of data at $1.07'$. So for δ below the former only the ellipse is compatible, in every syntax; with longitudes alone, only below $1.41'$. Compatibility alone thus eliminates the circle-based vicarious hypothesis once the data language contains distances: this reproduces the abandonment of circular orbits, not Kepler's last step from an oval to the ellipse, which is outside the pool.

The witness at $\tau_{\mathrm{d}} = 1\%$. The elements $e_1 = 0.102835781$, $e_2 = 0.0837969992$, $w = -1.33587144$, $L = -0.882335245$ (radians), $n = 0.00914611825$ radians per day and scale 1.52509743 fit every distance of the vicarious hypothesis to within 1.00000% and every longitude to within $5.119'$. The fit is sensitive to rounding: with the elements rounded to six decimals (the rounding of n to 0.009146 is the largest change) the longitude error rises to $6.2'$. The vicarious entry at 0.05% ($132.04'$) is the independent search's; the other search found $132.35'$ there.

Reading. Distances order the forms differently from longitudes. The bisected equant fits the distances to within 0.5% without giving up any longitude accuracy, and to 0.22% at best. The eccentric circle fits distances alone to 0.23%, but not with the elements its longitudes need: the distance fit wants a smaller eccentricity ($e = 0.142$ at 5%, against 0.185 for the longitude fit). The vicarious hypothesis fits distances alone to 0.016% (with $e_1 \approx e_2$ it is close to the equant), but not together with its longitude fit. The ellipse fits both at once, at its longitude minimax, for every τ_{d} down to 0.05%. Not computed: the Occam factors with distance data, and hence the MAP form above the thresholds; a lower bound on any least tolerance (it would need, for example, interval arithmetic over a covering of the element space); and the precision of Kepler's own distance determinations, so the table does not say which τ_{d} was historically available.

A.7 Reproducing the computations

The scripts in the accompanying repository are deterministic or seeded (seed 20261015 for the residual fits and the Mars pool, 20261016 for the graded-fit and distance computations) and were run with Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1 and pyerfa 2.0.1.5. Their saved outputs contain every computed number of Section 1 and of this appendix, except the two sampler figures (the bounding-box fraction and the number of chain steps), which follow from the script's parameters; the other figures are from the literature cited. Re-running reproduces them apart from timing lines; the Mars-pool posterior with the value of K3 takes about 80 minutes on one core, the distance search about 36 minutes, the residual fits about 2 minutes and the graded-fit computations under a minute. The independently written implementations quoted above are in the same repository.

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